

**BEACON: A Telemetry-Based Software Framework
for State-of-Health and Remaining-Useful-Life
Estimation of Electric-Vehicle Batteries**

A PROJECT REPORT

Submitted by

V NAVEEN

(CH.EN.U4CCE23054)

SANJAY P

(CH.EN.U4CCE23042)

in partial fulfillment for the award of the degree of

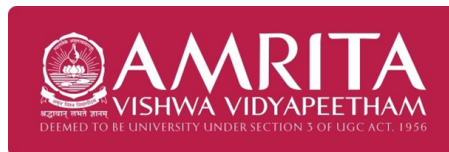
**BACHELOR OF TECHNOLOGY IN COMPUTER AND
COMMUNICATION ENGINEERING**

Under the guidance of

Dr. M GAYATHRI

Assistant Professor

Submitted to



**AMRITA SCHOOL OF ENGINEERING
AMRITA VISHWA VIDYAPEETHAM
CHENNAI – 601103**

October 2026

**BEACON: A Telemetry-Based Software Framework
for State-of-Health and Remaining-Useful-Life
Estimation of Electric-Vehicle Batteries**

A PROJECT REPORT

Submitted by

V NAVEEN

(CH.EN.U4CCE23054)

SANJAY P

(CH.EN.U4CCE23042)

in partial fulfillment for the award of the degree of

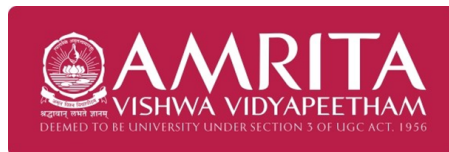
**BACHELOR OF TECHNOLOGY IN COMPUTER AND
COMMUNICATION ENGINEERING**

Under the guidance of

Dr. M GAYATHRI

Assistant Professor

Submitted to



**AMRITA SCHOOL OF ENGINEERING
AMRITA VISHWA VIDYAPEETHAM
CHENNAI – 601103**

October 2026



**SCHOOL OF
ENGINEERING
CHENNAI**

BONAFIDE CERTIFICATE

Certified that this project report “**BEACON: A Telemetry-Based Software Framework for State-of-Health and Remaining-Useful-Life Estimation of Electric-Vehicle Batteries**” is the bonafide work of **V NAVEEN (CH.EN.U4CCE23054)**, **SANJAY P (CH.EN.U4CCE23042)** who carried out the project work under my supervision.

SIGNATURE

Dr. LAKSHMANAN S A
CHAIRPERSON

Associate Professor
Department of ECE,
Amrita School of Engineering,
Amrita Vishwa Vidyapeetham, Chennai.

SIGNATURE

Dr. M GAYATHRI
SUPERVISOR

Assistant Professor
Department of ECE,
Amrita School of Engineering,
Amrita Vishwa Vidyapeetham, Chennai.

The Viva-Voce Examination for Project Phase-I was conducted on _____

INTERNAL EXAMINER

Name: _____

Designation: _____

EXTERNAL EXAMINER

Name: _____

Designation: _____



SCHOOL OF
ENGINEERING
CHENNAI

DECLARATION BY THE CANDIDATE

We declare that the report entitled **“BEACON: A Telemetry-Based Software Framework for State-of-Health and Remaining-Useful-Life Estimation of Electric-Vehicle Batteries”** submitted by us for the degree of BACHELOR OF TECHNOLOGY IN COMPUTER AND COMMUNICATION ENGINEERING is the record of the project work carried out by us under the guidance of **Dr. M GAYATHRI** and this work has not formed the basis for the award of any degree, diploma, associateship, fellowship, or other similar title in this or any other University or institution of higher learning.

SIGNATURE

V NAVEEN
CH.EN.U4CCE23054

SIGNATURE

SANJAY P
CH.EN.U4CCE23042

ABSTRACT

A battery management system (BMS) measures terminal voltage, current and temperature, but whether software above it can turn these into a trustworthy state of health (SOH) and remaining useful life (RUL) is usually answered by fitting a model. This project builds BEACON, a three-tier telemetry and health-monitoring system, and tests that question with measured evidence. BEACON ingests data from a CAN bus, a USB serial bench rig or research datasets, converges all three on one schema, and scores them through one shared pipeline that is served over a REST API to a React dashboard. On 31 NASA cells in nine protocol cohorts, eleven of twelve methods (all but a mean-only baseline) lost skill under leave-one-cohort-out validation, so the system's two working estimators fit nothing across cells. SOH is the ratio of charge delivered within a fixed voltage window to the same quantity on the cell's own first discharges, with ohmic sag corrected by a resistance identified from the voltage step at each load onset. From voltage, current and time alone, on 22 CALCE lithium cobalt oxide cells, the median per-cell error was 1.7% (95% CI 1.4–4.3%) on 17 cells, with five refused for stated causes, and the step resistance tracked the cyclers' resistance on 19 of 20 cells. Unchanged, the estimator measured all eight cells of an independent Oxford dataset at 3.0% median error. RUL extrapolated from each cell's own fade was within ± 20 cycles 73% of the time in the last 25 cycles before 90% capacity. The system refuses to output a number it cannot justify: every output is validated, banded by its measured error, or refused with a reason. A bench rig built on a NodeMCU ESP8266 with an INA219 and an LM35 streamed 83 of 83 checksummed frames at a measured 1.000 s cadence, which validates data acquisition but not health estimation.

Keywords: battery management system, state of health, remaining useful life, electric vehicle, internal resistance, telemetry, uncertainty quantification

ACKNOWLEDGEMENT

This project work would not have been possible without the contribution of many people. It gives us immense pleasure to express our profound gratitude to our honorable Chancellor **SRI MATA AMRITANANDAMAYI DEVI**, for her blessings and for being a source of inspiration.

We are indebted to extend our gratitude to our Director, **Mr. I B MANIKANTAN**, Amrita Vishwa Vidyapeetham, Chennai for facilitating us all the facilities and extended support to gain valuable education and learning experience.

We register our special thanks to **Dr. V. JAYAKUMAR**, Principal, Amrita Vishwa Vidyapeetham, Chennai for the support given to us in the successful conduct of this project. We wish to express our sincere gratitude to **Dr. LAKSHMANAN S A**, Chairperson, Department of Electronics and Communication Engineering, and our supervisor **Dr. M GAYATHRI** for their inspiring guidance, personal involvement, and constant encouragement during the entire course of this work.

We are grateful to Project Coordinator, Review Panel Members and the entire faculty of the Department of Electronic & Communication Engineering, for their constructive criticisms and valuable suggestions which have been a rich source to improve the quality of this work.

V NAVEEN
CH.EN.U4CCE23054

SANJAY P
CH.EN.U4CCE23042

TABLE OF CONTENTS

ABSTRACT	iv
ACKNOWLEDGEMENT	v
LIST OF TABLES	x
LIST OF FIGURES	xii
LIST OF SYMBOLS AND ABBREVIATIONS	xiv
1 INTRODUCTION	1
1.1 BACKGROUND	1
1.2 MOTIVATION	1
1.3 PROBLEM STATEMENT	2
1.4 RESEARCH GAP	2
1.5 OBJECTIVES	3
1.6 BATTERY HEALTH INDICATORS	3
1.6.1 State of Health	3
1.6.2 Internal Resistance	4
1.6.3 Remaining Useful Life	4
1.7 PROPOSED APPROACH	4
1.8 SIGNIFICANCE OF THE WORK	4
1.9 SCOPE AND ORGANISATION OF THE REPORT	5
2 LITERATURE REVIEW	6
2.1 STATE ESTIMATION IN THE BATTERY MANAGEMENT SYSTEM	6
2.2 DATA-DRIVEN SOH ESTIMATION AND ITS VALIDATION	6
2.3 DISCHARGE-CURVE ANALYSIS	7
2.4 REMAINING USEFUL LIFE PROGNOSTICS	7
2.5 PUBLIC DATASETS	7
2.6 TELEMETRY ACQUISITION IN VEHICLES AND LABORATORY RIGS	7
2.7 BATTERY AGEING MECHANISMS	8

2.8	TAXONOMY OF HEALTH ESTIMATION METHODS	8
2.9	INTERNAL RESISTANCE ESTIMATION IN THE LITERATURE	8
2.10	VEHICLE DATA INTERFACES	9
2.10.1	Controller Area Network	9
2.10.2	Serial Laboratory Rigs	9
2.10.3	Open Datasets	9
2.11	DIGITAL TWINS AND FLEET MONITORING	9
2.12	EXPLAINABILITY AND OUTPUT TRANSPARENCY	10
2.13	VALIDATION PRACTICE AND DATA LEAKAGE	10
2.14	COMPARISON OF APPROACH FAMILIES	10
2.15	IDENTIFIED RESEARCH GAPS	10
2.16	SUMMARY	11
3	METHODOLOGY / SYSTEM DESIGN	12
3.1	PROBLEM AND DESIGN OBJECTIVE	12
3.1.1	Research Variables	12
3.2	SYSTEM REQUIREMENTS	12
3.3	ASSUMPTIONS AND CONSTRAINTS	13
3.4	OVERALL SYSTEM FRAMEWORK	13
3.4.1	Unified Scoring Function	13
3.5	DATASETS	14
3.6	UNIFIED TELEMETRY SCHEMA	15
3.6.1	Derived Fields	15
3.6.2	Validation of the Schema	16
3.7	DATA FLOW AND PROCESSING SEQUENCE	16
3.8	DISCHARGE SEGMENTATION	17
3.9	VOLTAGE-WINDOW SOH ESTIMATION	17
3.10	INTERNAL RESISTANCE ESTIMATION AND OHMIC CORRECTION	17
3.11	DATA-QUALITY GATES AND REFUSAL CONDITIONS	18
3.12	LEARNED VOLTAGE WINDOWS FOR PARTIAL DISCHARGES	19
3.13	REMAINING USEFUL LIFE ESTIMATION FROM CELL-SPECIFIC FADE	19
3.14	EVIDENCE SUFFICIENCY CRITERION	19
3.15	HEURISTIC BEHAVIOUR SCORING LAYER	20
3.16	DIGITAL-TWIN STATE MODEL	20
3.17	DESIGN DECISION RECORDS	21
3.18	ALGORITHM SUMMARY	21
3.19	VALIDATION PROTOCOL	22
3.20	PARAMETERS OF THE ESTIMATORS	23

3.21	EVALUATION METRICS	23
3.22	EXPERIMENTAL CONTROLS AND REPRODUCIBILITY	24
4	IMPLEMENTATION AND DEPLOYMENT	25
4.1	IMPLEMENTATION OVERVIEW	25
4.2	DEVELOPMENT ENVIRONMENT AND TOOLS	25
4.3	CODE ORGANISATION	25
4.4	TELEMETRY ACQUISITION AND THE UNIFIED FRAME	26
4.4.1	Interface Between Acquisition and Scoring	26
4.5	UNIFIED SCHEMA IMPLEMENTATION	27
4.6	CAN AND DBC ADAPTER	28
4.6.1	Vehicle Registry	28
4.6.2	Limits of the Adapter	28
4.7	SERIAL WIRE PROTOCOL	28
4.7.1	Frame Delimitation	29
4.7.2	Error Detection	29
4.7.3	Channel Utilisation	29
4.8	IMPLEMENTATION OF REFUSAL GATES	29
4.9	HARDWARE BENCH RIG	31
4.9.1	Requirement and Design Calculations	31
4.9.2	Implemented Rig Configuration	31
4.9.3	Firmware	32
4.9.4	Bring-Up Procedure	33
4.10	SERVICE TIER: REST API, GATEWAY AND DASHBOARD	33
4.11	DIGITAL TWIN AND FLEET STORE	35
4.12	API REFERENCE	36
4.13	SECURITY CONSIDERATIONS	36
4.13.1	Threats Outside the Scope of the Design	37
4.14	DASHBOARD COMPONENTS	37
4.15	HEALTH REPORT CARD	38
4.16	TESTING, CONTINUOUS INTEGRATION AND REPRODUCIBILITY	38
4.16.1	Test Modules by Subsystem	39
4.17	CONTAINERISATION AND DEPLOYMENT	39
4.18	OPERATIONAL WORKFLOW	40
4.19	CHAPTER SUMMARY	40
5	RESULTS AND DISCUSSION	41
5.1	PERFORMANCE OF FITTED MODELS UNDER PROTOCOL SHIFT	41
5.2	FIELD STATE-OF-HEALTH ESTIMATION	42

5.3	PARTIAL-DISCHARGE SOH ESTIMATION	44
5.4	INTERNAL RESISTANCE ESTIMATION	44
5.5	REMAINING USEFUL LIFE ESTIMATION	45
5.6	RUL ESTIMATION UNDER SPARSE COMPLETE DISCHARGES	46
5.7	EVIDENCE SUFFICIENCY ANALYSIS	46
5.8	EXTERNAL VALIDATION ON THE OXFORD DATASET	47
5.9	CROSS-DATASET VALIDATION ON THE NASA DATASET	48
5.10	SYSTEM AND HARDWARE RESULTS	48
5.10.1	Bench Rig Captures	48
5.10.2	Fault Injection	49
5.10.3	Scope of the Hardware Results	49
5.11	CASE STUDY: HEALTH REPORT CARD	49
5.12	DISCUSSION	51
5.13	IMPLICATIONS FOR BMS SOFTWARE DESIGN	52
5.14	VALIDATION SUMMARY	52
5.15	ASSESSMENT AGAINST THE OBJECTIVES	53
5.16	THREATS TO VALIDITY	54
5.17	ROBUSTNESS AND SECURITY CHECKS	54
6	CONCLUSION AND FUTURE WORK	55
6.1	SUMMARY OF THE WORK	56
6.2	LIMITATIONS	56
6.3	FUTURE WORK	56
6.4	LESSONS LEARNED	57
6.5	RECOMMENDATIONS FOR DEPLOYMENT	57
	REFERENCES	58
	APPENDIX A	61
	APPENDIX B	62
	APPENDIX C	63

LIST OF TABLES

Table 1.1	Organisation of the Report	5
Table 2.1	Families of State-of-Health Estimation	8
Table 2.2	Validation Splits and Their Purpose	10
Table 2.3	Comparison of Approach Families	11
Table 2.4	Research Gaps and Corresponding Report Sections	11
Table 3.1	Functional and Non-Functional Requirements	13
Table 3.2	Data Sources and Their Role	14
Table 3.3	Core Columns of the Unified Schema	15
Table 3.4	Risk Categories of the Rule-Based Layer	20
Table 3.5	Health Index, Battery State and Digital-Twin State	20
Table 3.6	Architecture Decision Records	21
Table 3.7	Parameters Fixed Before External Validation	23
Table 4.1	Software Components Used	26
Table 4.2	Organisation of the Core Package	27
Table 4.3	Frame Size and Link Performance of the JSON Codec	29
Table 4.4	Refusal Gates and Their Purpose	30
Table 4.5	Shunt Sizing for the INA219 (320 mV Full Scale, 10 μ V LSB)	31
Table 4.6	Design Specification Versus Implemented Rig	32
Table 4.7	Firmware Compilation Record (<code>arduino-cli</code> , all warnings enabled)	32
Table 4.8	Services	33
Table 4.9	Principal Routes of the Python Service	36
Table 4.10	Security Review of the Service	37
Table 4.11	Views of the Dashboard	38
Table 4.12	Selected Test Modules and Verified Behaviour	39

Table 5.1	Benchmark on 31 NASA Cells in Nine Cohorts (Six of the Methods Shown); LOCO Intervals Are 95% Bootstrap Over Folds	41
Table 5.2	Feature Ablation and Shuffle Control (NASA SOH, LOCO)	42
Table 5.3	Field SOH at the 3.90–3.60 V Window; Median Per-Cell Absolute Error Against Cyclers Capacity, 95% Bootstrap Interval Over Cells	43
Table 5.4	SOH from Real Partial Cycling (CALCE CS2 Types 5 and 6)	44
Table 5.5	RUL to 90% Capacity. Left: By True Cycles to End of Life. Right: By Predicted Cycles, With the 90% Range of (True – Predicted)	45
Table 5.6	RUL Estimation from Sparse Complete Discharges (Methods Compared)	46
Table 5.7	Error by the Consistency Criterion of Eq. (3.7)	46
Table 5.8	External Validation on Eight Oxford Cells: Median and Worst Per-Cell Mean Absolute SOH Error	47
Table 5.9	Bench Rig Captures	48
Table 5.10	Status of the Main Outputs	53
Table 5.11	Objectives and Evidence	53
Table B.1	Artifacts and Scripts Behind Each Results Section	62

LIST OF FIGURES

Figure 3.1	Overall system framework: three transports, one unified frame, one scoring function, and the service tier.	14
Figure 3.2	Field SOH estimator: from raw telemetry to a banded or refused output.	18
Figure 4.1	A data frame: sentinel BEACON1, kind D, JSON body, and the XOR checksum *06, ending in a newline.	29
Figure 4.2	Implemented bench rig: measurement chain from the cell to the host.	32
Figure 4.3	Overview screen of the BEACON dashboard (light theme) on a synthetic fleet. The banner marks the values as simulated, and the panel at the bottom shows the measured bench-rig readings, with the health index and remaining life refused because no temperature channel exists.	34
Figure 4.4	Battery health screen (light theme): state of health over cycles for one battery, the attribution of its health index to behaviour terms, and its aggregate usage parameters (synthetic fleet).	35
Figure 5.1	LOBO and LOCO R^2 for SOH for every scored method on the 31-cell NASA benchmark (data: <code>benchmark_results.csv</code>). Skill measured across cells does not carry over to a new protocol.	42
Figure 5.2	SOH from voltage, current and time against every laboratory capacity test. Top: two constant-current cells, fixed window (mean error 0.9 and 1.1 points). Bottom: two cells of real partial cycling, learned window, estimate from partial discharges only (3.4 and 3.2 points).	43
Figure 5.3	Mean absolute SOH error for every measured CALCE cell, field method; the dashed line is the median (1.7%).	44
Figure 5.4	RUL to 90% capacity. Left: predicted against actual cycles left in the last 50 cycles. Right: share of estimates within ± 20 cycles by actual distance to end of life.	45

- Figure 5.5 Stage B capture from the bench rig (83 frames over 82 s, one cell at rest). Voltage moves between 3.984 and 3.992 V in steps of the INA219's 4 mV LSB, and current stays at the ± 0.4 mA noise floor. 49
- Figure 5.6 Evidence view of the BEACON dashboard (light theme) for the bench rig: 83 of 83 records accepted, voltage and current as measured, temperature reported as not supplied, and the health index, remaining life and state of health refused because coverage is incomplete. 50
- Figure 5.7 Abridged health report card for CS2_35 at cycle 90, with the laboratory check. 50

LIST OF SYMBOLS AND ABBREVIATIONS

ADR	- Architecture Decision Record
API	- Application Programming Interface
BMS	- Battery Management System
CAN	- Controller Area Network
CC/CV	- Constant Current / Constant Voltage
CI	- Confidence Interval (or Continuous Integration, by context)
DBC	- CAN database file describing signal layout
EV	- Electric Vehicle
LCO	- Lithium Cobalt Oxide
LOBO	- Leave-One-Battery (cell)-Out validation
LOCO	- Leave-One-Cohort-Out validation
LRC	- Longitudinal Redundancy Check (XOR checksum)
MAE	- Mean Absolute Error
NMC	- Lithium Nickel Manganese Cobalt Oxide
OCV	- Open-Circuit Voltage
RUL	- Remaining Useful Life
SOC	- State of Charge
SOH	- State of Health
R_{step}	- Apparent resistance identified from the load step